1 Cantor's Theorem

Cantor's work into the theory of infinite sets extends far beyond the conclusions of Theorem 1.5.6. Although initially resisted, his creative and relentless assault in this area eventually produced a revolution in set theory and a paradigm shift in the way mathematicians came to understand the infinite.

Cantor's Diagonalization Method

Cantor published his discovery that $\mathbb{R}$ is uncountable in 1874. Although it has some modern polish on it, the argument presented in Theorem 1.5.6(ii) is actually quite similar to the one Cantor originally found. In 1891, Cantor offered another proof of this same fact that is startling in its simplicity. It relies on decimal representations for real numbers, which we will accept and use without any formal definitions.

Theorem 1.1 (Theorem 1.6.1)**.** *The open interval* $(0,1) = \{x \in \mathbb{R} : 0 < x < 1\}$ *is uncountable.*

Problem 1.2. Show that $(0,1)$ is uncountable if and only if $\mathbb{R}$ is uncountable. This shows that Theorem 1.6.1 is equivalent to Theorem 1.5.6.

Proof: ($\rightarrow$) Assume that $(0,1)$ is uncountable. Assume for the sake of contradiction that $\mathbb{R}$ is countable. Then we know by **Theorem 1.5.7** that since $(0,1) \subseteq \mathbb{R}$, then $(0,1)$ must be either countable, finite, or empty. This contradicts our assumption that $(0,1)$ is uncountable. Thus $\mathbb{R}$ must be uncountable.

($\leftarrow$) Assume $\mathbb{R}$ is uncountable. Let $f : (0,1) \rightarrow \mathbb{R}$ where $f(x) = \tan(\pi(x - \frac{1}{2}))$. We know that $f(x)$ is $(1-1)$ and onto. Therefore, since we have a bijective function from $(0,1)$ to $\mathbb{R}$, then we know $(0,1)$ is uncountable.

□

Continuing the proof for **Theorem 1.1**:

Proof: As with Theorem 1.5.6, we proceed by contradiction and assume that there exists a function $f : \mathbb{N} \rightarrow (0,1)$ that is 1–1 and onto. For each $m \in \mathbb{N}$, $f(m)$ is a real number between 0 and 1, and we represent it using the decimal notation

$$f(m) = .a_{m1}a_{m2}a_{m3}a_{m4}a_{m5}\ldots$$

What is meant here is that for each $m, n \in \mathbb{N}$, the digit $a_{mn} \in \{0, \ldots, 9\}$ is the nth digit in the decimal expansion of $f(m)$.

We arrange this correspondence in the array:

$\mathbb{N}$	$(0,1)$
1	$f(1) = .a_{11}a_{12}a_{13}a_{14}a_{15}\ldots$
2	$f(2) = .a_{21}a_{22}a_{23}a_{24}a_{25}\ldots$
3	$f(3) = .a_{31}a_{32}a_{33}a_{34}a_{35}\ldots$
4	$f(4) = .a_{41}a_{42}a_{43}a_{44}a_{45}\ldots$
$\vdots$	$\vdots$

Now for the key idea. We define a real number $x \in (0,1)$ with decimal expansion

$$x = .b_1b_2b_3b_4\ldots$$

using the rule

$$b_n = \begin{cases} 2 & \text{if } a_{nn} \neq 2, \\ 3 & \text{if } a_{nn} = 2. \end{cases}$$

To compute b_1, look at a_{11}. If $a_{11} = 2$, set $b_1 = 3$; otherwise, set $b_1 = 2$.

Problem 1.3.

(a) Explain why $x = .b_1b_2b_3b_4\ldots$ cannot be $f(1)$.

Solution: x and $f(1)$ differ at the first decimal place.

(b) Explain why $x \neq f(2)$, and in general why $x \neq f(n)$ for any $n \in \mathbb{N}$.

Solution: x and $f(2)$ differ at the second decimal place. For any $f(n)$, $n \in \mathbb{N}$, x and $f(n)$ differ from each other at the nth decimal place.

(c) Point out the contradiction that arises from these observations and conclude that $(0,1)$ is uncountable.

Since we know that x and $f(n)$ differ from each other at the nth decimal place, then we have produced a real number $x \in \mathbb{R}$ that is not contained in our list. Therefore, this contradicts the fact that we assumed our list contains all of $\mathbb{R}$ and that our function $f : (0,1) \to \mathbb{R}$ is $(1-1)$ and onto. Therefore $\mathbb{R}$ is uncountable.

□

Discussion: The main catch-up, and the heart of the proof, is that we assume that our list contains all of $\mathbb{R}$. In order to show that $\mathbb{R}$ is uncountable, we need to find an element of $\mathbb{R}$ that is not in this list. We do this by rigging this new number to be different than any number on this list by "flipping" the value of the decimal place that falls on the main diagonal. Because this number differs from every element in the list at the nth decimal place, there is no way it can be in the list to begin with. Therefore, we can't enumerate $\mathbb{R}$, then $\mathbb{R}$ must be uncountable.

Problem 1.4. Provide rebuttals to the following complaints about the proof of Theorem 1.6.1.

(a) Every rational number has a decimal expansion, so the same diagonal argument applied to rationals would "prove" the rationals are uncountable. Because we know any subset of $\mathbb{Q}$ is countable, the proof must be flawed.

(b) Some numbers have two decimal representations. For instance, $1/2 = .5 = .4999\ldots$. Does this cause a problem?

Solution:

(a) We know that $\mathbb{Q}$ is countable and therefore can be listed.This argument requires that you produce an honest list of all the elements in $\mathbb{Q}$. So, if we show that a rational number is not in a list then the list was not complete. In addition to this reasoning, rational numbers either terminate or have an infinite pattern that repeats. The denationalization argument creates an irrational number which is not in $\mathbb{Q}$, so there is no contradiction. We already knew that this new irrational number was not in the list of $\mathbb{Q}$.

(b) This could cause a subtle problem, but in our example, we do not use 0 or 9. This avoids this issue, so decimal ambiguity does not spoil the argument.

□

Problem 1.5. Let S be the set of all sequences of 0's and 1's:

$$S = \{(a_1, a_2, a_3, \ldots) : a_n = 0 \text{ or } 1\}.$$

Give a rigorous argument showing that S is uncountable.

Proof. Assume, for the sake of contradiction, that S is countable. This means that we can list its elements:

$$S^1 = (a_{11}, a_{12}, a_{13}, \ldots)$$

and

$$S^2 = (a_{21}, a_{22}, a_{23}, \ldots)$$

and

$$\ldots$$

We can define a new sequence

$$t = (b_1, b_2, b_2, \ldots)$$

where $b_n = 1 - a_{nn}$ (this is just flipping the 1 or 0 in the a_{nn} spot in the list). Therefore $b_n \neq a_{nn}$ for each n, which contradicts our assumption that S is countable. Therefore, S is uncountable. □

Discussion: Why does this work? This works because we are producing a new irrational term of 1's and 0's. We are able to have this irrational number in our set S because we defined it to be any combination of 1's and 0's. If you put a decimal in front of the first number then you would pretty much have the exact argument made for the proof that $[0, 1]$ is uncountable. Because of this, you can also show that S has the same cardinality as $\mathbb{R}$.

Power Sets and Cantor's Theorem

Given a set A, the power set $\mathscr{P}(A)$ is the set of all subsets of A. It is important to understand that $\mathscr{P}(A)$ is itself a set whose elements are subsets of A.

Problem 1.6.

(a) Let $A = \{a, b, c\}$. List the eight elements of $\mathscr{P}(A)$. The Eight elements of $\mathscr{P}(A)$ are all possible subsets of the set A, so $\mathscr{P}(A) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

(b) If A is finite with n elements, show that $\mathscr{P}(A)$ has 2^n elements.

Proof: Going through the list of elements $A = \{a_1, a_2, ..., a_n\}$, we have the choice to either include or exclude each $a \in A$ to form our subset. This gives us the following total amount of possible choices or subsets:

$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdots = 2^n$$

Therefore A finite set A with n elements has a power set which contains 2^n elements.

□

Problem 1.7.

(a) Using $A = \{a, b, c\}$, construct two different 1–1 mappings from A into $\mathscr{P}(A)$.

(b) Let $C = \{1, 2, 3, 4\}$. Give an example of a 1–1 map $g : C \to \mathscr{P}(C)$.

(c) Explain why it is impossible in these cases to construct mappings that are onto.

Proof. (a) The singleton map works as one example. This is $f(x) = \{x\}$. Another example would be $f(x) = \{x, y\}$.

(b) Again, we can use the singleton map $f(x) = \{x\}$.

(c) In part (a), $|A| = 3$ but $|P(A)| = 8$. We can have at most 3 distinct elements in this case, which means that the function cannot cover the 8 total elements in the powerset. The same reasoning goes for part b where $|C| = 4$ and $|P(C)| = 16$. In general, the set will always have a size n and the power set will always have the size 2^n. So for infinite sets with $n \geq 1$, we will always have $|P(X)| = 2^n > n = |X|$.

□

Theorem 1.8 (1.6.2 (Cantor's Theorem))**.** *Given any set A, there does not exist a function $f : A \to \mathscr{P}(A)$ that is onto.*

Proof. This proof, like the others of its kind, is indirect. Thus, assume, for contradiction, that $f : A \to \mathscr{P}(A)$ is onto. Unlike the usual situation in which we have sets of numbers for the domain and range, f is a correspondence between a set and its power set. For each element $a \in A$, $f(a)$ is a particular subset of A.

The assumption that f is onto means that every subset of A appears as $f(a)$ for some $a \in A$. To arrive at a contradiction, we will produce a subset $B \subseteq A$ that is not equal to $f(a)$ for any $a \in A$.

Construct B using the following rule. For each element $a \in A$, consider the subset $f(a)$. This subset of A may contain the element a or it may not. This depends on the function f. If $f(a)$ does not contain a, then we include a in our set B. More precisely, let

$$B = \{a \in A : a \notin f(a)\}.$$

Problem 1.9 (1.6.7)**.** Return to the particular functions constructed in Exercise 1.6.6 and construct the subset B that results using the preceding rule. In each case, note that B is not in the range of the function used.

Solution:

(a) $f(x) = \{x\} \implies f(a) = \{a\}, \quad f(b) = \{b\}, \quad f(c) = \{c\}$. This means $a \notin B$, $b \notin B$, and $c \notin B$, so $B = \varnothing$. The range of f is $\{\{a\}, \{b\}, \{c\}\}$ and $\varnothing$ is not in this range so B is also not in the range of f.

(b) $f(x) = \{x, y\} \implies f(a) = \{a, b\}$, $f(b) = \{b, c\}$, $f(c) = \{a, c\}$. Therefore $a, b, c \notin B$ making $B = \varnothing$, and again B is not in the range of f. So in both examples, B is not in the range.

□

We now focus on the general argument. Because we have assumed that our function $f : A \to \mathscr{P}(A)$ is onto, it must be that $B = f(a_0)$ for some $a_0 \in A$. The contradiction arises when we consider whether or not a_0 is an element of B.

Problem 1.10.

(a) Show that the assumption $a^* \in B$ yields a contradiction.

Case 1: Assuming $a' \in B$, then we know by definition of B, that $a' \notin f(a')$ for some $a' \in A$. Thus $a' \notin B$, which is a contradiction.

(b) Show that the assumption $a^* \notin B$ also yields a contradiction.

Case 2: Assuming $a' \notin B$ we then know $a' \in f(a')$. If $a' \in f(a')$, we know once again $B = f(a')$, so therefore $a' \in B$, which is a contradiction.

Therefore, there exists no mapping $f : A \to \mathscr{P}(A)$ that is onto.

Discussion: We have once again uniquely defined a set B that forces the contradiction in both cases. By defining $B = \{a \in A : a \notin f(a)\}$, we can use the definition of a function being onto to guarantee that there exists an $a' \in A$ such that $f(a') = B$. Examining both of these cases, we see that this function cannot be onto because then our definition of B results in mathematical nonsense. Arriving at a contradiction in both cases, we know there exists no onto mapping from a set to its power set.

Problem 1.11. Using the tools developed so far, give a compelling argument that

$$\mathscr{P}(\mathbb{N}) \sim \mathbb{R}.$$

Proof: In order to prove $\mathscr{P}(\mathbb{N}) \sim \mathbb{R}$, we will need to find a function from $\mathscr{P}(\mathbb{N})$ to $\mathbb{R}$ that is bijective. We will first accomplish this by representing each subset of the natural numbers as an infinite binary expansion. Going through the natural numbers, we define X as a specific subset of $\mathbb{N}$ and a specific $x \in X$, where $x = .n_1n_2n_3\ldots$ is defined for if $n_m \in \mathbb{N}$ is in the subset or not with a 1 if it is in the subset or a 0 if it is not.

For example, the expansion $.101010\ldots$ would tell us that all of the odd numbers are in our specific subset.

Now that we can represent every possible subset as a binary sequence, we can use the fact that each real number in $(0, 1)$ has an infinite binary expansion. It should be stated we will only include one binary expansion for each unique real number, much like we do in the proof of **Theorem 1.1** (for $.9999\cdots = 1$ if we dont use 2 or 3) . This gives us a way to bijectively map all possible subsets to the set $(0, 1)$, which we know also bijectively maps to $\mathbb{R}$. Thus $\mathscr{P}(\mathbb{N}) \sim \mathbb{R}$.

Discussion: This proof works because we know that creating these binary sequences will give us both a way to represent each possible subset of the natural numbers and a way to represent the real numbers in $(0, 1)$ (It is important this proccess goes both ways). Since we are ignoring the case where two binary sequences are actually the same real number, much like $.999999999\cdots = 1$, we can perform this mapping to from all subsets to the interval $(0, 1)$ in a $(1 - 1)$ and onto fashion. Then since we know $(0, 1)$ bijectively maps to $\mathbb{R}$ we know $\mathscr{P}(\mathbb{N}) \sim \mathbb{R}$.

Problem 1.12. Answer each of the following by establishing a 1–1 correspondence with a set of known cardinality.

(a) Is the set of all functions from $\{0, 1\}$ to $\mathbb{N}$ countable or uncountable?

Proof: The set of all functions from $\{0, 1\}$ to $\mathbb{N}$ is countable. We can represent all possible functions that take $\{0, 1\}$ to $\mathbb{N}$ as pair $(f(0), f(1)) \in \mathbb{N} \times \mathbb{N}$. We get the following list:

	$f(1) = 0$	$f(1) = 1$	$f(1) = 2$	$f(1) = 3$ $\cdots$
$f(0) = 0$	$(0, 0)$	$(0, 1)$	$(0, 2)$	$(0, 3)$ $\cdots$
$f(0) = 1$	$(1, 0)$	$(1, 1)$	$(1, 2)$	$(1, 3)$ $\cdots$
$f(0) = 2$	$(2, 0)$	$(2, 1)$	$(2, 2)$	$(2, 3)$ $\cdots$
$f(0) = 3$	$(3, 0)$	$(3, 1)$	$(3, 2)$	$(3, 3)$ $\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Where we can list all elements (m, n) in a new set A_p with the rule that $m + n = p$. Therefore, we can enumerate the set of all possible functions from $\{0, 1\}$ to $\mathbb{N}$ and

therefore the set of all functions from $\{0,1\}$ to $\mathbb{N}$ is countable.

□

Discussion: If you look back to classwork from 1.5 (roughly the second week of notes in September), we actually previously showed $\mathbb{N} \times \mathbb{N}$ is countable

(b) Is the set of all functions from $\mathbb{N}$ to $\{0,1\}$ countable or uncountable?

Proof: The set of all functions from $\mathbb{N}$ to $\{0,1\}$ is uncountable. Following Cantor's Diagonalization argument, we will assume the set of all functions from $\mathbb{N}$ to $\{0,1\}$ is countable. We know for all $f(n)$, where $n \in \mathbb{N}$, that $f(n)$ will map a natural number to either 0 or 1. We can show all possible functions with the natural numbers as inputs in the list below.

	1	2	3	4	5	
f_1	1	0	1	1	0	$\cdots$
f_2	0	1	1	0	1	$\cdots$
f_3	1	1	0	0	1	$\cdots$
f_4	0	0	1	1	1	$\cdots$
f_5	1	0	0	1	0	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

More formally we have

	1	2	3	4	5	
f_1	$f_1(1)$	$f_1(2)$	$f_1(3)$	$f_1(4)$	$f_1(5)$	$\cdots$
f_2	$f_2(1)$	$f_2(2)$	$f_2(3)$	$f_2(4)$	$f_2(5)$	$\cdots$
f_3	$f_3(1)$	$f_3(2)$	$f_3(3)$	$f_3(4)$	$f_3(5)$	$\cdots$
f_4	$f_4(1)$	$f_4(2)$	$f_4(3)$	$f_4(4)$	$f_4(5)$	$\cdots$
f_5	$f_5(1)$	$f_5(2)$	$f_5(3)$	$f_5(4)$	$f_5(5)$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Then, since we know that this is just going to be an arbitrary sequence of binary strings, we can apply the rule on the diagonal of our list where we can create a new function f_α where:

$$f_\alpha = \begin{cases} 1 & \text{if } f_n(n) \neq 1, \\ 0 & \text{if } f_n(n) = 1. \end{cases}$$

Then we know f_α cannot be on the list because it will differ from all other functions at the nth index. This contradicts that our assumption that the list of all functions from $\mathbb{N}$ to $\{0,1\}$ can be enumerated. Thus the set of all functions from $\mathbb{N}$ to $\{0,1\}$ is uncountable.

□

Discussion: This proof follows almost step for step with Exercise 1.5, just differs in how everything is flipped (these methods are still equivalent). Realistically, this proof could

be a lot shorter by just saying once you represent the function and its natural number inputs, this is the same as the set of all arbitrary strings of 0s and 1s for Exercise 1.5.

2 Section 1.7

The following is a short recap of the discussion we during our presentation. This is a short section to read through and goes through the extensive impact that Cantor and his diagonalization argument has had on mathematics.

Cantor's demonstration that there are multiple infinite cardinalities (countable infinity vs. continuum) contradicted a long-held intuition. It showed that cardinality is a rich and structured notion: $|\mathbb{N}| < |\mathbb{R}|$, and moreover $|\mathscr{P}(A)| > |A|$ for any set A. Cantor's diagonal argument offered a distinct method to separate sizes of infinite sets concretely. Diagonalization has appeared in many other contexts. One of them is in logic and computability like in Turing's halting problem or Godel's incompleteness theorems.

This work on sets and the infinite brought up the question if there could be a set strictly between $|\mathbb{N}|$ and $|\mathbb{R}|$. This is called the Continuum Hypothesis. Godel and Cohen showed that the Continuum Hypothesis can neither be proved nor disproved, and it does not matter if you choose it to be true or false, because there will be no contradictions anywhere else in math.

Another interesting idea that came up from this topic is that you cannot have a single largest infinity. This means that there are an infinite number of infinities. So, the phrase "Let U be the set of all possible things" is paradoxical because card $U <$ card $P(U)$, which would mean that U did not contain everything. Therefore, you can not have a largest infinity. An easy way to get an increasing size of infinity is by just continuously taking the power set of a set. Each power set is larger than the last and this pattern goes on forever. This idea of a set not being able to contain everything is where Cantor thought he disproved God because by this, God cannot be all possible things.